

LINKING WEALTH AND PUNISHMENT EFFECTIVENESS: PUNISHMENT AND COOPERATION UNDER CONGRUENT HETEROGENEITIES

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Global and local cooperation in supplying global public goods is often insufficient. In this respect, laboratory experiments show that peer punishment is an effective cooperation-enhancing instrument. However, it is unclear whether peer punishment would facilitate cooperation and public good provision even under congruent heterogeneities in wealth and punishment effectiveness. To this end, we experimentally study the effect of peer punishment under joint heterogeneities, where either the richest or the poorest member is also the most effective punisher. We compare these joint heterogeneities to treatments with single heterogeneities in either endowment or punishment effectiveness and to a baseline symmetry treatment with homogeneous parties. We find that heterogeneity in punishment effectiveness does not matter for cooperation, whereas endowment heterogeneity reduces cooperation compared to symmetry. This is because rich members contribute a lower portion of their endowment to the public good than their poorer counterparts. We also observe that cooperation is higher under joint heterogeneities in endowment and punishment effectiveness than under endowment heterogeneity (with no differences than under symmetry). This holds even when the rich party gains less from cooperation and is the most effective punisher. (JEL C92, D74, H41)

I. INTRODUCTION

International cooperation in supplying global public goods is often insufficient (e.g., preventing climate change, sustaining biodiversity, preventing epidemics, and fighting terrorism). Even though some international bodies have been established for this purpose (Intergovernmental Panel on Climate Change, World Health Organization, World Trade Organization, the United Nations Counter-Terrorism Implementation Task Force, etc.), they usually cannot enforce compliance, so more often than not they fail to attain sufficient public good provision.¹ This failure has inspired the scientific community to identify

and test “bottom-up” cooperation-enhancing institutions that do not require a government or a ruling body. One strand of this literature is dedicated to the use of experimental methods to study stylized versions of this dilemma, for example, Isaac and Walker (1988), Cadsby and Maynes (1999), Ostrom, Walker, and Gardner (1992), Milinski, Semmann, and Krambeck (2002), Ahn, Isaac, and Salmon (2009), and Friedman and Oprea (2012). In this respect, one institution that has been shown to be especially effective in facilitating and maintaining cooperation in the experimental laboratory is (costly) peer punishment (Ostrom, Walker, and Gardner 1992; Yamagishi 1986; and especially Fehr and

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1. For example, in the context of human rights, Hafner-Burton and Tsutsui (2005, 1378) write that “[...] global human

rights treaties supply weak institutional mechanisms to monitor and enforce regime norms, offering governments strong incentives to ratify human rights treaties as a matter of window dressing rather than a serious commitment to implement respect for human rights in practice.” Regarding climate change, Jacquet and Jamieson (2016, 645) write that “[t]he Paris Agreement ultimately rests more on economic, social and political obligation than it does on legal authority.”

ABBREVIATION

ECU: Experimental Currency Units

Gächter 2000, 2002). The performance of this institution is remarkable in two ways. First, costly punishment is viewed as a “second-order public good,” and in a finitely repeated game, the subgame perfect equilibrium indicates that a money-maximizing individual would neither punish nor cooperate. Second, it does not require a central government or an agreed ruling body and hence could be potentially used for the provision of (global) public goods among sovereign states. The idea of sanctions is not new in this context. Trade sanctions were used in banning chlorofluorocarbons causing ozone depletion, and there is a growing consensus that they should be included in climate change negotiations (e.g., Barrett 1997, 2001; Cai, Riezman, and Whalley 2013; Sunstein 2007; Tian and Whalley 2010; Tingley and Tomz 2014). Also in this regard, Newnham (2008, 188–192) explains how the United States threatened nations into joining the coalition of the willing for the invasion of Iraq in 2003.

The findings of Fehr and Gächter (2000, 2002) showing that peer punishment is very effective in facilitating cooperation in a public good game has triggered a broad literature concentrating on the generalizability of this institution in establishing cooperation (e.g., a meta-analysis by Balliet, Mulder, and Van Lange 2011; and a review by Chaudhuri 2011). One characteristic shared by virtually all field situations (including global and local domains) is that some parties are wealthier than others, and some can punish more severely than others.² Consequently, few studies have investigated the effect of either endowment heterogeneity (Prediger 2011; Reuben and Riedl 2013; Visser and Burns 2015; Weng and Carlsson 2015) or heterogeneity in punishment effectiveness (Dorrough, Glöckner, and Lee 2017; Nikiforakis, Normann, and Wallace 2010). The overall findings of these studies indicate that peer punishment is no less effective in maintaining cooperation under either endowment heterogeneity or heterogeneity in the effectiveness of punishment (see details in Section II).

However, one remarkable gap in the literature is that although heterogeneities in the field are typically observed in congruence, that is, rich parties can punish more harshly than their poor counterparts, the effect of such congruent heterogeneities has so far not been investigated in the

laboratory. This situation is especially intriguing when the rich benefit less from the public good than their poorer counterparts. In such a case it is hard to imagine how “poor and weak punishers” could impose cooperation on their “rich and effective punishers” counterparts. Our overarching motivation in initiating this research is to study whether or not the remarkable performance of the peer punishment institution in public good provision also extends to the arguably realistic condition of congruent heterogeneities in wealth and punishment effectiveness, where the rich party is also the most effective punisher. This characterizes (and arguably still characterizes) the global domain where the United States is the richest and strongest country. Moreover, even the skeptical reader, who is less inclined to find similarities between global public good provision and a simple public good game experiment, would hardly deny that congruent heterogeneities also characterize public good provision at the local level (e.g., in contributing to building a bridge, digging water wells, or funding a school in a small village).³

Bearing this in mind, we report here on a laboratory experiment designed to fill the gap in the literature by studying the effect of congruent heterogeneities in wealth and punishment effectiveness on public good provision. We study two polar cases of joint heterogeneities: “*rich strong*,” where the richest party is the most effective punisher, and “*poor strong*,” where the poorest party is the most effective punisher. We study the “poor strong” case as a counterfactual to the “rich strong” case with congruent heterogeneities. In particular, in a linear public good game, the potential gain of the poor from cooperation is higher than that of the rich, and the “poor strong” case, where the poor are the most effective punishers and can force the rich into cooperation, seems to be more promising to enhance cooperation than any joint heterogeneity treatment.⁴ To disentangle the effects of the joint heterogeneities, we compare these

3. These examples are taken from Carlsson, Johansson-Stenman, and Nam (2014) and Miguel and Gugerty (2005) (although the focus of these studies is not on congruent heterogeneities). Moreover, even in a fairly egalitarian society such as the Teop, a small-scale society from Papua New Guinea, the authorities (“big men”) are chosen precisely because they own land (more precisely, in this matrilineal society, “big men” are chosen because their mothers own land) (see Grimalda, Pondorfer, and Tracer 2016).

4. Concrete examples for the “poor strong” case are not as easy to come up with as they are for the “rich strong” case. However, in the global domain, powerful positions in

2. There are also other important heterogeneities between parties, for instance, in returns from the public good (e.g., Nikiforakis, Noussair, and Wilkening 2012; Reuben and Riedl 2009, 2013).

“poor strong” and “rich strong” treatments to two treatments with a single heterogeneity, either in endowment or in effectiveness of punishment, and to a baseline *symmetry* treatment with homogeneous parties.

Our findings can be summarized as follows: Heterogeneity in punishment effectiveness has no bearing on cooperation or punishment. By contrast, heterogeneity in endowment appears to reduce cooperation. This happens because, under endowment heterogeneity, rich members contribute a lower portion of their endowment than poor members (and a lower portion than rich members in the joint heterogeneity treatments). Further, joint heterogeneities in endowment and punishment effectiveness (either when the rich are the most effective punishers or when the poor are the most effective punishers) do not reduce cooperation in comparison with symmetry. This result holds true even when the rich party gains less from cooperation and is the most effective punisher. Moreover, there is no difference in cooperation between the two joint heterogeneity treatments. However, cooperation requires less punishment under congruent heterogeneities (where the richest party is also the most effective punisher) than in the counterfactual case (in which the poorest party is the most effective punisher). This indicates that congruent heterogeneities in wealth and punishment power could be rather beneficial for public good provision.

II. RELATED LITERATURE

In the following, we summarize the literature on the effect of heterogeneities in either endowments or punishment effectiveness on cooperation in linear public good games.⁵ In

the United Nations Security Council rotate among nonpermanent member states. Currently, among the 15 Security Council members are developing countries such as Bolivia, Côte d'Ivoire, Equatorial Guinea, and Ethiopia. On a local (and possibly more intuitive) level, “poor strong” situations arise when the poor have little to lose (e.g., from an uprising). We are grateful for an anonymous referee for coming up with this point.

5. The effect of endowment heterogeneity was also investigated in other social dilemma games, for example, threshold public good games (e.g., Bagnoli and McKee 1991; Rapoport and Suleiman 1993), nonlinear public good games (e.g., Chan et al. 1996, 1999), common pool resource games (e.g., Cardenas 2003), etc. However, the theoretical predictions of these games are different from the prediction of the linear public good game (where the Nash equilibrium and the socially efficient outcomes are at the boundaries of the choice set). We, therefore, refrain from summarizing their findings.

particular, we review studies on the effect of (1) endowment heterogeneity in the absence of the possibility of peer punishment, (2) endowment heterogeneity when the parties can punish each other, and (3) heterogeneity in punishment effectiveness.

We start with heterogeneity in endowments in the absence of the possibility to punish group members. In a meta-analysis of linear public good game experiments, Zelmer (2003) observes that endowment heterogeneity negatively affects contributions to the public good. Cherry, Kroll, and Shogren (2005) find that overall contributions are lower under endowment heterogeneity than under homogeneous endowments. Buckley and Croson (2006) find that poor (i.e., low-endowment) members contribute similar amounts to the public good as their rich (i.e., high-endowment) counterparts. This implies that the rate of cooperation (i.e., share of contribution from the endowment) is higher for the poor than that for the rich members. Hofmeyr, Burns, and Visser (2007) find that endowment heterogeneity has no significant impact on contributions to the public good, and that the rate of cooperation is similar between the rich and poor members. Anderson, Mellor, and Milyo (2008) observe that when a fixed (show-up) payment is publicly announced, heterogeneity in such fixed payments reduces contributions to the public goods. Moreover, Keser et al. (2013) find that contributions are not different between homogeneous and “weakly heterogeneous” endowments. However, contributions to the public good are reduced under “strongly heterogeneous” endowments, where a rich member is endowed with substantially more money than his/her counterparts. In line with Buckley and Croson (2006), they also observe that rich members contribute similar amounts to their poor counterparts, and hence their rate of cooperation is lower than that of their poorer counterparts. Reuben and Riedl (2013) find that when peer punishment is not allowed, there are little differences in contributions between rich and poor members and the differences vanish over time. This is explained by deterioration in contributions of both rich and poor members over time toward zero.⁶ Weng and Carlsson (2015) report on a considerably lower average cooperation under

6. The treatment closest to ours is with endowment heterogeneity and unrestricted contributions. Notably, Reuben and Riedl (2013) focus on contribution norms under heterogeneities, so they do not compare endowment heterogeneity and heterogeneity for a given group endowment.

endowment heterogeneity.⁷ Finally, Hargreaves Heap, Ramalingam, and Stoddard (2016) show that endowment heterogeneity reduces contributions to the public good. This happens because under endowment heterogeneity the rich members contribute less than when endowed the same amount but under endowment homogeneity.

Now we turn to the effect of heterogeneity in endowment when punishment is allowed.⁸ Reuben and Riedl (2013) find that when punishment is allowed and there is no deterioration in contributions over time, the rich members contribute more than their poor counterparts. Yet, cooperation rates are slightly higher by the poor than by the rich. Furthermore, Reuben and Riedl (2013) do not identify differences in punishment inflicted by poor and rich members. In a study examining participants from fishing communities in South Africa, Visser and Burns (2015) find that endowment heterogeneity leads to higher overall contributions to the public good than endowment homogeneity. They also observe that cooperation rates are slightly higher by the poor than by the rich, and that members of heterogeneous groups are less likely to punish if the average group contribution is high. In addition, in an experiment with rural dwellers from Southern Namibia, Prediger (2011) finds that endowment heterogeneity leads to higher contributions (and less punishments) than endowment homogeneity (with and without punishment). Finally, in a study that focuses on the interaction between group identity, punishment, and endowment distribution in contribution to the public good (Weng and Carlsson 2015), there appears to be no significant pure effect of endowment heterogeneity on contributions or punishments.⁹

7. This study hones in on the interaction between group identity, punishment, and endowment distribution in contributions to the public good. The average cooperation rate in the “Homo” treatment is 0.49 while only 0.31 in the “Hetero” treatment. Under strong identity manipulation, cooperation increases in the heterogeneous endowment treatment but not in the homogeneous endowment treatment (see Weng and Carlsson 2015, 30, Table 2).

8. Notably, De Cremer and van Dijk (2009) also study punishment and endowment heterogeneity. However, their main focus is on contributions of members for the implementation of a sanctioning system, and their design is not close enough to allow a comparison with the studies reviewed in this section (drawing on Fehr and Gächter’s design).

9. The average cooperation rate is 0.69 under endowment homogeneity and 0.67 under endowment heterogeneity. The average punishment is 0.53 under endowment homogeneity and 0.47 under endowment heterogeneity. In “pure effect,” we refer to the “Homo-Punish1:3” and “Hetero-Punish1:3” treatments (without group identity manipulation). Under strong identity manipulation, average cooperation rate

The effect of heterogeneity on the effectiveness of punishment was first studied by Niki-forakis, Normann, and Wallace (2010). They compare the case of homogeneous players to the case where one player has a higher *punishment ratio* (punishee’s fine divided by punisher’s fee) than the remaining three group members. These authors find that heterogeneity in the effectiveness of punishment does not induce lower contributions than under symmetry. Additionally, they do not observe any significant differences in contributions between (heterogeneous) group members. In a recent study, Dorrough, Glöckner, and Lee (2017) find that heterogeneity in punishment (one member with a higher punishment ratio than his/her peers) does not per se reduce contributions. However, contributions are reduced when punishment heterogeneity is combined with instability (i.e., when members compete in each period to be assigned the highest punishment ratio in a subsequent period).

In sum, when *peer punishment is not allowed*, most studies indicate that endowment heterogeneity reduces contributions to the public good. However, when *peer punishment is allowed* evidence is ambiguous (the findings are either that heterogeneity does not matter or that it rather increases contributions). Finally, previous studies indicate that heterogeneity in punishment effectiveness neither increases nor reduces contributions in comparison with symmetry. In the following, we study the effect of joint heterogeneities in endowment and punishment effectiveness on contributions to the public good. To our knowledge this has not been investigated before.

III. DESIGN, CONJECTURES, AND PROCEDURE

Our workhorse is a linear public good game with punishment à la Fehr and Gächter (2000, 2002). At the beginning of the experiment, participants are randomly assigned to fixed groups of three members (partner matching). The experiment lasts exactly 20 periods, each consists of two stages. In the first stage, each participant decides how to divide up an initial endowment, $E_i \in \{10, 20, 30\}$, nominated in experimental currency units (hereafter ECU) between keeping it and investing it in a project that yields 0.6 ECU to every group member for each 1 ECU invested. In

is higher under endowment homogeneity than under endowment heterogeneity (0.79 vs. 0.67). See Weng and Carlsson (2015, 30, Table 2).

the second stage, participants are informed about the contributions of each of their group members and can assign negative points to each of the other two members. Each negative point assigned to a group member costs the punisher one ECU and reduces the first-stage payoff of the punishee by the amount of $F_i \in \{1, 2, 3\}$. Note that because the punishment cost is equal to one for all members, F_i also denotes the *punishment ratio* (ratio of the fine for the punishee divided by the fee for the punisher). The payoff of a participant i in any given period is as follows¹⁰:

$$(1) \quad \Pi_i = E_i - C_i + 0.6 \sum_{n=\sum j+i} C_n - \sum_{j \neq i} F_j P_{ji} - \sum_{i \neq j} P_{ij},$$

where E_i denotes the initial endowment, C_i the participant's contribution to the public good, and $\sum_n C_n$ the sum of the contributions of the three group members to the public good. In addition, $\sum_{j \neq i} F_j P_{ji}$ indicates the total payoff reduction from being punished by the other two members (depending on each member's punishment ratio (F_j) and on the number of punishment points she has inflicted (P_{ji})). Finally, $\sum_{j \neq i} P_{ij}$ denotes the participant's cost from punishing other members. At the end of the experiment, participants are paid the sum of their payoffs from each of the 20 periods plus the 25 ECU initially allotted to every group member.¹¹

Our design is shown in Table 1. We conducted the following five treatments (keeping the average endowment and punishment ratio at 20 ECU and 2, respectively). (1) *Symmetry* (Sym) with individual endowments of 20 ECU and a punishment ratio equal to 2. (2) *Asymmetric punishment* (Asym P) with individual endowments of 20 ECU and punishment ratios equal to 3, 2, and 1. (3) *Asymmetric endowment* (Asym E) with individual endowments of 30, 20 and 10 ECU, and a punishment ratio equal to 2. (4) *Poor strong* with

individual endowments of 30, 20 and 10 ECU and respective punishment ratios of 1, 2, and 3. (5) *Rich strong* with individual endowments of 30, 20, and 10 ECU and respective punishment ratios of 3, 2, and 1.

Two design features are worth emphasizing. First, the three punishment ratio levels are chosen following Nikiforakis and Normann (2008), who find that changes in punishment ratio between 1, 2, and 3 have considerable effects on cooperation (they yield 33%, 57%, and 87% cooperation, respectively).¹² Second, under endowment heterogeneity, the maximum gains from cooperation (if all members fully cooperate) of the rich (who are endowed with 30 ECU), medium (20 ECU), and poor (10 ECU) members are 6, 16, and 26 ECU (i.e., 20%, 80%, and 260% of the endowments, respectively). The maximum losses (if the respective member is the only one to fully cooperate, while the other two members free ride fully on his/her cooperation) are 12, 8, and 4 ECU for the rich, medium, and poor parties (in relative terms, exactly 40% of the endowment for the three parties). Thus, full cooperation (i.e., contributing the whole endowments to the public good), although in the interest of all members, is less beneficial in absolute and relative terms and riskier in absolute terms for the rich members than for their poorer counterparts.

A. Ex-Ante Conjectures

We now formulate the possible conjectures as to the effects on cooperation (defined as rate of contribution from the total endowment) of the joint heterogeneities in wealth and punishment effectiveness.¹³ From Equation (1), it is easy to verify that an individual with standard (money maximizing) preferences would neither punish nor cooperate in any of the (finitely repeated) treatments. Against this prediction, peer punishment has been consistently shown to yield stable cooperation in public good games. This result could be explained using models of social preferences yielding a continuum of cooperative

10. In line with Fehr and Gächter (2002), punishments cannot reduce more than the first (contribution) stage payoff, and a punishee cannot receive more than a total of 10 punishment points. Moreover, following the standard procedure, punishment is anonymous (i.e., punishees are only informed about the overall punishment they have received, not about who has punished them).

11. The initial amount of 25 ECU follows Fehr and Gächter (2000). Participants are allotted this initial amount to prevent bankruptcy if the punishment costs exceed total payoff. In addition, all participants who came to the experiment received a show-up fee of EUR 3.

12. Related results are by Anderson and Putterman (2006), Carpenter (2007), and Égas and Riedl (2008).

13. Fudenberg and Pathak (2010) show that punishment enhances cooperation irrespective of whether punishees are aware of being punished. Hence, it is not only actual punishment but also deterrence that increases cooperation (see also Gächter, Renner, and Sefton 2008 showing that while punishment is reduced over time, cooperation is maintained through deterrence). Therefore we refrain from formulating hypotheses about punishment levels, focusing instead on cooperation (irrespective of whether it is established through actual punishment or deterrence).

TABLE 1
Design Parameters and Classifications to Member Types

Treatment	Endowment (E_i)	Punishment Ratio (F_i)	Classification I (E_i)	Classification II (F_i)
Symmetry (24 groups)	20	2	Medium	Medium
	20	2	Medium	Medium
	20	2	Medium	Medium
Asymmetric punishment (25 groups)	20	3		Strong
	20	2		Medium
	20	1		Weak
Asymmetric endowment (27 groups)	30	2	Rich	
	20	2	Medium	
	10	2	Poor	
Poor strong (28 groups)	30	1	Rich	Weak
	20	2	Medium	Medium
	10	3	Poor	Strong
Rich strong (32 groups)	30	3	Rich	Strong
	20	2	Medium	Medium
	10	1	Poor	Weak

Nash equilibria (depending on the specific model and parameters). In light of the variety of models (and parameters), we follow previous studies on heterogeneities (including those studies reviewed in Section II) in refraining from relying on a parametric prediction of a specific model for our a priori conjectures. Instead, we base our conjectures (where possible) on evidence from previous studies.

Nikiforakis, Normann, and Wallace (2010) acknowledge that, on the one hand, heterogeneity in the punishment ratio may solve the coordination problem as to who should punish a defector when the most effective punisher perceives his/her role as a norm enforcer (i.e., to punish defectors). This would increase cooperation. On the other hand, a punishment that depends heavily on one member is frail and may be less effective in facilitating cooperation. It may even be the case that the effective punisher would exploit his/her power to free ride on his group members' contributions. A priori, it is unclear which of these contradicting effects is stronger. Moreover, both Nikiforakis, Normann, and Wallace (2010) and Dorrough, Glöckner, and Lee (2017) find that punishment heterogeneity has no significant effect on cooperation. Therefore our first conjecture is formulated as follows:

Ho(1): *Heterogeneity in punishment effectiveness in the Asymmetric punishment treatment will not yield different levels of cooperation than in the Symmetry treatment.*

Endowment heterogeneity may also have two potentially contrasting effects. On the one hand, under such heterogeneity no contribution leads to differences in final payoffs between

group members, which may result in envious low-endowment (poor) members punishing their richer counterparts.¹⁴ On the other hand, under endowment heterogeneity the potential gain of the high-endowment (rich) members is lower than that of their poorer counterparts, and their potential loss from cooperation is higher. Hence, the rich may be less keen to cooperate than under homogeneous endowment. The empirical evidence in those studies with peer punishment is also ambiguous. Visser and Burns (2015) and Prediger (2011) find that endowment heterogeneity increases overall cooperation, while Weng and Carlsson (2015) observe no such effect. In the absence of clear evidence, our second conjecture is formulated as follows:

Ho(2): *Heterogeneity in endowment in the Asymmetric endowment treatment will not yield different levels of cooperation than in the Symmetry treatment.*

Now we turn to the novelty of the study, the two joint heterogeneity treatments. Note that the considerations formulated for endowment heterogeneity also hold in the joint heterogeneity treatments. However, in *Poor strong*, the poor members (who gain more from cooperation) are more effective punishers than their rich counterparts and are thus expected to force the rich into cooperation. Hence,

Ho(3): *We expect higher cooperation levels in the Poor strong treatment than in the Asymmetric endowment treatment.*

14. This is predicted for example by inequity aversion preferences (Fehr and Schmidt 1999). In fact, it may be sufficient to assume that a fraction of the (poor) participants have such preferences to induce the rich to cooperate.

Next let us consider perhaps the most realistic scenario, where heterogeneities in wealth and punishment are congruent (*Rich strong*). On the one hand, the punishment threat of poor members (who gain more from cooperation than the rich) is now considerably lower than that of their rich counterparts. This might reduce cooperation in comparison with *Asymmetric endowment*. On the other hand, because of the high punishment effectiveness of the rich, it is less likely that their poorer counterparts (who gain more from the public good) would defect. So the rich may feel less vulnerable, and this could increase cooperation in comparison with *Asymmetric endowment*. As we cannot determine a priori which effect is stronger, our next conjecture reads as follows:

Ho(4): *Congruent heterogeneities in the Rich strong treatment will not yield different levels of cooperation than in the Asymmetric endowment treatment.*

Finally, a corollary of Ho(3) and Ho(4) leads to the following:

Ho(5): *We expect higher cooperation levels in the Poor strong treatment than in the Rich strong treatment.*

B. Procedure

A total of 408 students from different faculties and departments of a large German university participated in the experiment (52% female, 36% economics students, average age: 23.1). We conducted five sessions in the *Symmetry*, *Asymmetric punishment*, and *Asymmetric endowment* treatments, and six sessions in each of the novel *Rich strong* and *Poor strong* treatments. In sum, there are 24, 25, 27, 28, and 32 independent groups in the *Symmetry*, *Asymmetric punishment*, *Asymmetric endowment*, *Poor strong*, and *Rich strong* treatments, respectively. Participants were recruited using Hroot (Bock, Baetge, and Nicklisch 2014), and the experiment was programmed and conducted using *z-Tree* software (Fischbacher 2007). It lasted about 60 minutes. The average earning per participant was EUR 21.50 (including EUR 3 show-up fee).

IV. RESULTS

We start by inspecting contributions and punishment behavior across treatments.¹⁵ Then we

15. For comparability, we only refer to punishment point inflicted ($\sum P_{ij}$) or received ($\sum P_{ji}$) but not to the overall

look into the behavior of the different member types across treatments. Throughout this part of the analysis, we focus on behavior in periods 10–18, after participants had gained considerable experience but before end-game effect set in (unlike in a one-shot game, individual actions within groups are interrelated over time, so we treat the individual decisions per group as independent observations). For comparison across treatments we use the robust rank order test (Feltovich 2003).¹⁶ Finally, we use (parametric) econometric models to study individual cooperation and punishment behavior.

A. Aggregate Behavior

Table 2 shows summary statistics of the contributions, punishment received, and payoffs in the different treatments for both the initial period and the second half of the experiment (periods 10–18). In addition, Figure 1 illustrates the evolution of average contributions (Figure 1A) and the rate of groups attaining full cooperation (contribute all their endowment to the public good) (Figure 1B). Table 2 and Figure 1A indicate that (1) cooperation (i.e., rate of contribution from the total endowment) in the initial period is highest under *Symmetry*¹⁷ and (2) over the second half of the experiment, average cooperation is persistently about 10%–20% lower in *Asymmetric endowment* than in the other treatments. Formally, using a robust rank order test for our conjectures, we find that in periods 10–18 cooperation is significantly lower under *Asymmetric endowment* than under *Symmetry* ($p < .01$).¹⁸ By

impact of the punishment (point received times punishment ratio).

16. In particular, for the analysis in period 1, we treat individual decisions as independent observations. However, individuals' decisions starting from period 2 are influenced also by past decisions of their group members, so for the analysis over periods 10–18 we treat independent observations at the group level. Finally, in the robust rank order test, we report on the one-sided p values (i.e., increase in cooperation vs. no change in cooperation).

17. Regarding first-period contributions, the significance levels are of $p < .04$ (one-sided robust rank order test) in comparing *Symmetry* with any of the heterogeneity treatments. For the analysis of the initial period, we use individuals' cooperation levels as independent observations (a total of 72, 75, 81, 84, and 96 individuals in the *Symmetry*, *Asymmetric punishment*, *Asymmetric endowment*, *Poor strong*, and *Rich strong* treatments, respectively).

18. For the analysis over periods 10–18, we use averaged individual contributions per group as independent observations (there are 24, 25, 27, 28, and 32 groups in the *Symmetry*, *Asymmetric punishment*, *Asymmetric endowment*, *Poor strong*, and *Rich strong* treatments, respectively).

TABLE 2
Average (SD) of Individual Contributions, Punishments Received, and Payoffs (per Group)

Treatment	Contributions		Punishment Received		Payoffs	
	Period 1	Periods 10–18	Period 1	Periods 10–18	Period 1	Periods 10–18
Symmetry	13.12 (3.45)	17.23 (4.16)	0.64 (1.05)	0.43 (1.26)	28.58 (4.45)	32.48 (5.45)
Asymmetric punishment	11.08 (3.57)	15.35 (6.32)	1.23 (1.97)	0.42 (0.79)	24.85 (7.68)	31.04 (5.86)
Asymmetric endowment	11.17 (3.32)	13.55 (5.29)	0.63 (0.98)	0.53 (0.64)	27.05 (3.76)	29.25 (5.12)
Poor strong	11.33 (2.81)	16.09 (4.56)	1.21 (1.36)	0.87 (1.03)	25.36 (5.66)	30.22 (5.18)
Rich strong	11.80 (4.02)	15.65 (5.47)	0.85 (1.42)	0.41 (0.68)	26.57 (5.35)	31.22 (5.99)

contrast, cooperation is not different between *Symmetry* and *Asymmetric punishment*. As to the effect of the joint heterogeneities, we observe lower cooperation in *Asymmetric endowment* than in both *Poor strong* and *Rich strong* (each at $p = .03$). This result (lowest cooperation in *Asymmetric endowment*) is especially apparent when we look at the percentage of groups per treatment that manage to fully cooperate (i.e., group members contribute all their endowments, a total of 60 ECU per group). Figure 1B illustrates that in the second half of the experiment 59%, 45%, 38%, and 47% of the groups in *Symmetry*, *Asymmetric punishment*, *Poor strong*, and *Rich strong*, respectively, manage to fully cooperate in periods 10–18. By contrast, in *Asymmetric endowment* only 21% of the groups manage to do so.¹⁹ Turning to punishment (see Figure 2), we find that it is most conspicuous in *Poor strong*. Regarding the comparison of joint heterogeneities, we find significantly more punishment in *Poor strong* than in *Rich strong* ($p = .02$). We can now state our first result²⁰:

RESULT 1 (AGGREGATE COOPERATION AND PUNISHMENT): *In the second half of the*

19. These figures are of rates of full cooperation in each of periods 10–18. Alternatively, 42%, 32%, 32%, and 34% of the group in the *Symmetry*, *Asymmetric punishment*, *Poor strong*, and *Rich strong* manage to fully cooperate in all these (10–18) periods, but only 15% in *Asymmetric endowment*.

20. Further support for the following finding is provided in Appendix S1 (Supporting Information). Table A.1 shows the results of regression models (for periods 10–18, using groups as independent observations) for treatments effects. These models show that *Asymmetric endowment* yields lower cooperation and payoffs than *Symmetry* and that *Poor strong* and (marginally) *Asymmetric endowment* produce more punishment than *Symmetry*.

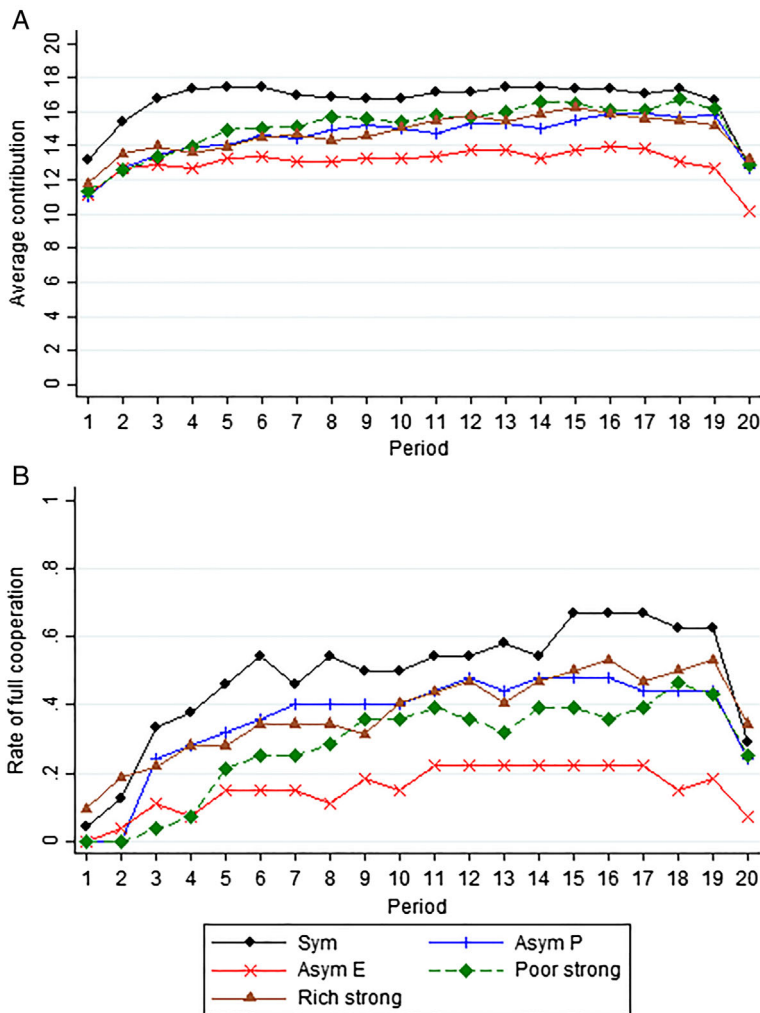
experiment, cooperation is lower in the Asymmetric endowment treatment than in the other treatments. There is more punishment in the Asymmetric endowment and Poor strong treatments than in the other treatments.

Let us now consider overall efficiencies due to the heterogeneities in endowment and punishment effectiveness. Efficiency increases with cooperation and decreases with punishment. It is captured by the individual payoffs (defined above by Equation (1)), so in the following we use individual payoff per group (see Table 2) as a measure of overall efficiency. In the second half of the experiment (periods 10–18), we find lower payoffs in *Asymmetric endowment* than under *Symmetry* ($p < .01$, robust rank order test) but no payoff differences between *Symmetry* and *Asymmetric punishment*. Turning to the joint heterogeneities, we observe lower payoffs in *Asymmetric endowment* than in *Rich strong*, but no payoff differences between *Asymmetric endowment* and *Poor strong* (or between the two joint heterogeneity treatments). Due to excessive punishment in *Poor strong*, it seems plausible that for those groups achieving high cooperation, efficiency in *Poor strong* is lower than in *Rich strong*. To determine whether this is the case, we now inspect only those groups that managed to achieve high cooperation (groups contributing on average 75% or more of the total endowment over periods 10–18).²¹ For these groups, we find that payoffs are lower in both *Asymmetric endowment* and *Poor strong* than in *Rich strong*

21. This amounts to 79%, 68%, 52%, 68%, and 66% of the groups in *Symmetry*, *Asymmetric punishment*, *Asymmetric endowment*, *Poor strong*, and *Rich strong*, respectively.

FIGURE 1

Evolution of Average Contributions and Rate of Full Cooperation. (A) Contributions, (B) Rate of Groups Attaining Full Cooperation



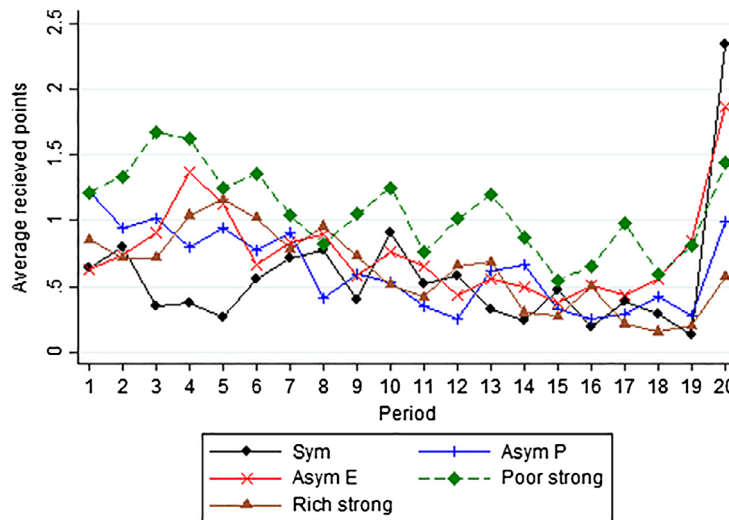
($p = .02$ and $.045$), but no significant difference can be found between the two former treatments. A possible interpretation is that for the case where peer punishment is successful in facilitating cooperation, congruent heterogeneities appear to increase efficiency in comparison with the case where the rich member is the least effective punisher.

B. Behavior by Member Type

We now relate the differences observed across treatments to the behavior of member types

varying in their endowments and punishment effectiveness. To this end, we define *rich* (*poor*) types as those members who are endowed with 30 (10) ECU in the treatments with endowment heterogeneity (i.e., *Asymmetric endowment*, *Poor strong*, and *Rich strong*). Similarly, across the heterogeneity in punishment effectiveness dimension (in *Asymmetric punishment*, *Poor strong*, and *Rich strong*), a *strong* (*weak*) *punisher* is defined as a member with a punishment ratio of 3 (1). The medium-type members are those who are endowed with 20 ECU and have

FIGURE 2
Evolution of Average Punishment



a punishment ratio of 2. Table 1 shows the classification according to types.

Figure 3 shows the evolution of cooperation by type in the different treatments. It illustrates that the reason for the lowest cooperation in *Asymmetric endowment* is the lower cooperation rate of its rich members. Formally, the rich-type members in this treatment cooperate less than the rich members in both *Poor strong* and *Rich strong* (both at $p = .02$). Moreover, only in *Asymmetric endowment* do we find significantly lower cooperation by the rich than by the poor ($p = .03$).²² Note that in all our treatments rich members contribute higher amounts than their poor and medium counterparts. Even in *Asymmetric endowment* the rich members contribute on average 19.30 ECU (i.e., 64% of their endowment), the medium type 13.44 (67% of their endowment), while the poor 7.91 (79% of their endowment). Turning to punishment, Figure 4 shows average punishment inflicted and received over periods 10–18. The rich members in *Asymmetric endowment* receive more punishment than the rich (and strong) in *Rich strong* ($p = .04$). Also, the rich (and weak) in *Poor strong* receive more punishment than the rich (and strong) in *Rich strong* ($p < .01$). As expected, the poor (and strong) in *Poor strong* inflict more punishment

than poor members in *Asymmetric endowment* ($p = .02$) and poor (and weak) members in *Rich strong* ($p < .01$). Hence, we can summarize our findings as follows:

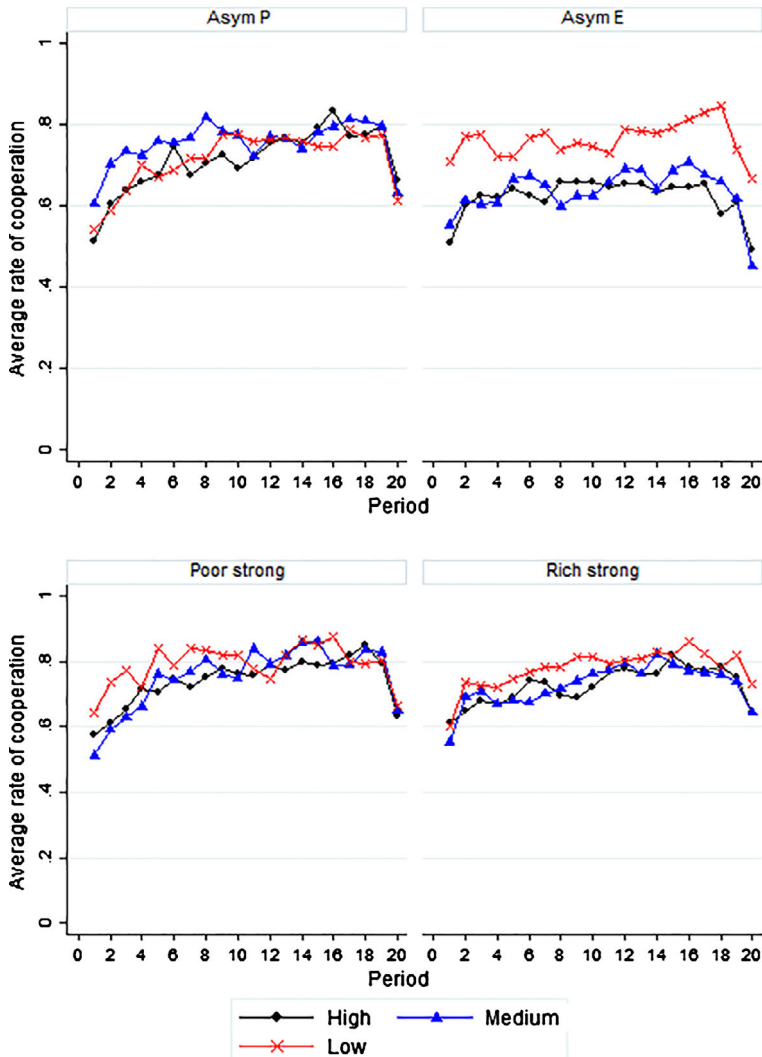
RESULT 2 (AGGREGATE COOPERATION AND PUNISHMENT BY TYPE):

In the second half of the experiment, the rich members in the Asymmetric endowment treatment cooperate less than their poor counterparts and also the rich members in the two joint heterogeneity treatments. The poor (and strong) members in the Poor strong treatment inflict more punishment on their group members than their poor (and weak) counterparts in the Rich strong treatment. Consequently, the rich members in the Asymmetric endowment treatment and the rich (and weak) members in the Poor strong treatment receive more punishment than the rich (and strong) members in the Rich strong treatment.

Results 1 and 2 suggest that the link between actual punishment and cooperation is moderated by the heterogeneities in endowment and punishment effectiveness. Although cooperation is indistinguishable between *Rich strong* and *Poor strong*, in the latter treatment it is achieved by more punishment. Interestingly, the differences in punishment behavior between the two joint heterogeneity treatments do not merely evolve over time but are already observed in the first period, suggesting that the channels leading to cooperation in these two treatments are fundamentally different.

22. In this treatment, the medium type members cooperate marginally less than the poor ($p = .09$). There is no difference in cooperation between the rich and medium types

FIGURE 3
Evolution of Cooperation across Heterogeneous Types



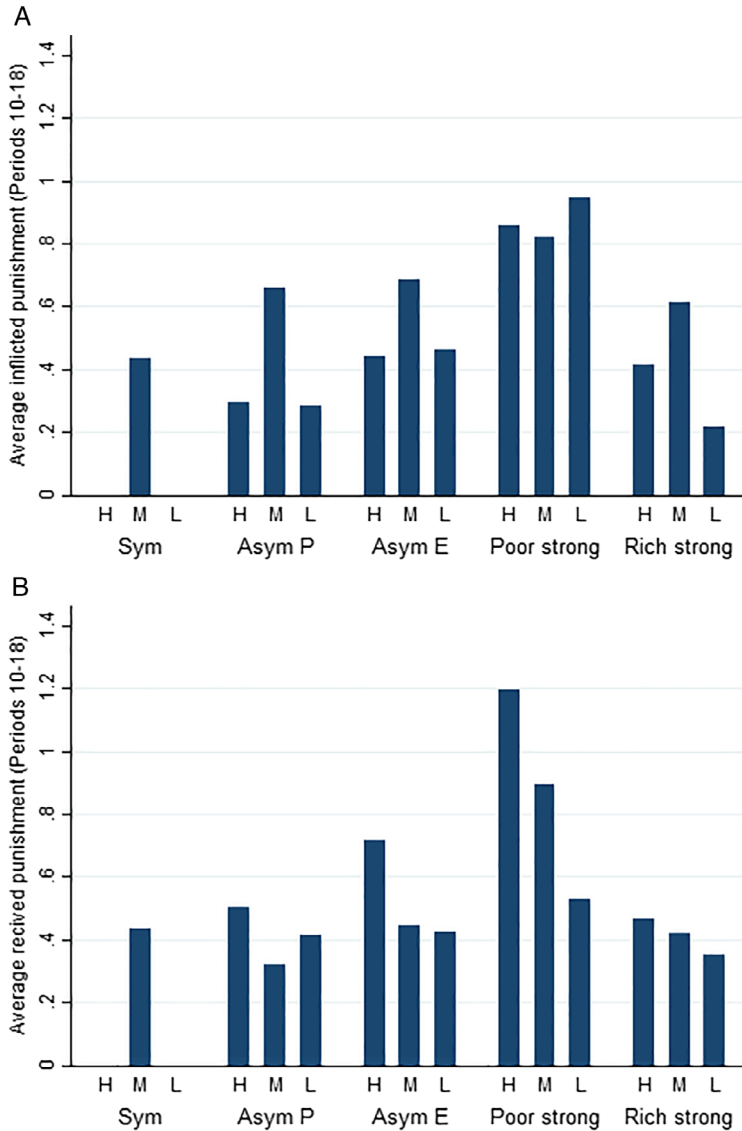
Notes: “High” (“Low”) denotes rich (poor) type with an endowment of 30 (10) ECU, except for the Asym P treatment where “High” (“Low”) stands for “strong” (“weak”) type with a punishment ratio of 3 (1). “Medium” denotes members with an endowment of 20 ECU and a punishment ratio of 2.

Furthermore, Reuben and Riedl (2013) consider two dimensions that guide contributions: *efficiency* and *reciprocity*. Efficiency requires that each member fully cooperates (i.e., contributes all of his/her endowment to the public good). As for reciprocity, under endowment heterogeneity there are three pertinent rules (under endowment homogeneity these rules converge): *equal contributions*,

equal earnings, and *equal cooperation*. Equal contributions require all members to contribute the same amount. Equal earnings demand that all members contribute, such that payoffs of all members are equalized. Finally, equal cooperation requires members to contribute proportionally to their endowments. Note, however, that full cooperation (i.e., contributing the whole endowments)

FIGURE 4

Punishment per Type in Periods (10–18). (A) Punishment Inflicted and (B) Punishment Received



Notes: “H” (“L”) denotes rich (poor) type with an endowment of 30 (10) ECU, except for the Asym P treatment where “H” (“L”) stands for “strong” (“weak”) type with a punishment ratio of 3 (1). “M” denotes members with an endowment of 20 ECU and a punishment ratio of 2.

satisfies the efficiency dimension as well as two reciprocity rules, equal cooperation and equal earnings.

Figure 3 indicates that in all treatments, the different member types contribute, on average, between 60% and 80% of the endowment to the public good. Due to the high cooperation rates,

payoffs are not very different between types (as full cooperation yields equal earnings). Hence, we cannot infer whether participants are driven by efficiency consideration or by norms of equality of cooperation or earnings. However, in line with the related literature on endowment heterogeneity (e.g., Buckley and Croson 2006) we

can distinguish between two reciprocity rules, equal contributions and equal cooperation. We find that in all treatments, except for *Asymmetric endowment*, contribution behavior is very much in line with equal cooperation (no differences in cooperation between types). Moreover, even in *Asymmetric endowment*, contribution behavior is closer to equal cooperation than to equal contributions. This could be due to the large difference in endowments between the member types; the endowment of the rich is three times that of their poor counterparts and 1.5 times that of the medium types.²³

C. Individual Decisions on Cooperation and Punishment

The analysis of the aggregate data can only reveal differences in cooperation and punishment levels separately but cannot account for the interdependences between cooperation and punishment. In the following, we use parametric model estimations to study individual behavior in adjustments of contributions and in punishment received.

We start by following previous studies (e.g., Fuster and Meier 2010) in regressing the *individual cooperation adjustment* ("change in the individual's cooperation between periods t and $t-1$ ") on the *lagged group cooperation adjustment* ("average change in cooperation of the other two group members between periods $t-1$ and $t-2$ ") and on *received punishment in period $t-1$* (indicating the payoff reduction from being punished in the previous period, that is, negative points received *times* punishment ratio of the punisher).²⁴

The regression results are shown in Table 3 and can be summarized as follows. First, in a model including all treatments, we find that an increase in group cooperation leads to a (future) increase in individual's cooperation. Received punishment leads to an increase in individual's cooperation in all treatments except for *Rich strong*.²⁵ Next, we study cooperation adjustment behavior across the two heterogeneity dimensions. We use the classification for

types displayed in Table 1 to estimate models for rich and poor members in the treatments with endowment heterogeneity (i.e., *Asymmetric endowment*, *Poor strong*, and *Rich strong*) and for strong and weak punisher members in the treatments with punishment heterogeneity (i.e., *Asymmetric punishment*, *Poor strong*, and *Rich strong*). In all these estimations (except for the strong punishers in *Asymmetric punishment*), we find that an increase in the other group members' cooperation leads to an increase in the individual's cooperation.²⁶ Turning to the effect of punishment received in a previous period, it increases cooperation for the rich (and weak) member in *Poor strong* ($p = .02$, F -test) but does not affect poor members in any of the treatments.²⁷ In comparing across treatments with punishment heterogeneity, punishment received in a previous period significantly increases strong punishers' cooperation only in *Asymmetric punishment* ($p < .01$), while for the weak punisher members it only increases cooperation when they are also rich (in *Poor strong*, $p = .02$).

RESULT 3 (INDIVIDUAL'S COOPERATION ADJUSTMENT PER TYPE): For virtually all types (rich, poor; strong, weak) an increase in the group members' cooperation leads to an increase in the individual's cooperation. Moreover, receiving a punishment increases cooperation by strong punisher members in the *Asymmetric punishment* treatment and by rich and weak punisher members in the *Poor strong* treatment.

The finding that the types in all treatments adjust their cooperation in response to changes in group member cooperation is in line with findings on conditional cooperation (e.g., Fischbacher and Gächter 2010; Fischbacher, Gächter, and Fehr 2001).

Next, we attempt to understand punishment behavior (i.e., "what affects receiving a punishment?"). Following Fehr and Gächter (2000) and subsequent studies, we use a tobit model to regress *punishment received* on *average levels of cooperation by the other two group members* and also on *negative deviation from this average*. For comparability across treatments and types, we only measure negative points received (i.e., $\sum P_j$), not the overall payoff

23. Table S2 in Appendix S1 shows average (SD) contributions and cooperation according to the member types.

24. We also estimated respective random effects panel data models. The results of the pooled regressions and of the panel data models are similar.

25. Using a joint-significance F -test, we find that group members' cooperation adjustment is significant at $p < .01$ in all treatments and that received punishment is significant in all treatments (at $p < .02$ or less) except for *Rich strong*.

26. At a significance level of $p < .04$ (but for the rich members in *Asymmetric endowment* with $p = .051$).

27. Punishment received also marginally increases cooperation for the rich in *Asymmetric endowment* ($p = .052$).

TABLE 3
Ordinary Least Squares Model Estimation (Change in Cooperation)

Treatments Members	ALL	With Endowment Heterogeneity (AE, PS, RS)		With Punishment Heterogeneity (AP, PS, RS)	
	(All Types)	(Rich Type)	(Poor Type)	(Strong Type)	(Weak Type)
Change in average cooperation by others ($t - 1$)	0.28*** (0.06)	0.18* (0.09)	0.26** (0.12)	0.23 (0.16)	0.36*** (0.07)
Change in average cooperation by others ($t - 1$) $\times$ AP	0.04 (0.07)	-	-	-	-
Change in average cooperation by others ($t - 1$) $\times$ AE	-0.11 (0.08)	-	-	-	-
Change in average cooperation by others ($t - 1$) $\times$ PS	-0.00 (0.08)	0.08 (0.13)	-0.01 (0.17)	0.01 (0.20)	-0.10 (0.12)
Change in average cooperation by others ($t - 1$) $\times$ RS	-0.11 (0.08)	0.11 (0.15)	0.01 (0.16)	0.06 (0.20)	-0.09 (0.12)
Punishment received ($t - 1$)	0.01*** (0.00)	0.01* (0.00)	0.01 (0.01)	0.01*** (0.00)	0.00 (0.00)
Punishment received ($t - 1$) $\times$ AP	-0.00 (0.00)	-	-	-	-
Punishment received ($t - 1$) $\times$ AE	-0.00 (0.01)	-	-	-	-
Punishment received ($t - 1$) $\times$ PS	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.01)	-0.00 (0.01)	0.00 (0.00)
Punishment received ($t - 1$) $\times$ RS	-0.01** (0.00)	-0.00 (0.00)	-0.01 (0.01)	-0.01 (0.01)	-0.00 (0.00)
Constant	-0.15*** (0.02)	-0.14*** (0.04)	-0.11*** (0.03)	-0.12*** (0.04)	-0.14*** (0.04)
Treatment dummies	Yes	Yes	Yes	Yes	Yes
Period dummies	Yes	Yes	Yes	Yes	Yes
Observations	7,344	1,566	1,566	1,530	1,530
F($k - 1, N - k - 1$)	9.57***	3.76***	2.55***	3.93***	6.31***
R ²	.09	.09	.07	.07	.12

Notes: Dep. variable: "Change in a participant i 's cooperation between periods t and $t - 1$." The table shows the estimated coefficients (standard errors clustered across groups in parentheses). The variable "Change in average cooperation by others ($t - 1$)" is defined as the difference in the average cooperation rate of the other two group members between periods $t - 1$ and $t - 2$. AP, AE, PS, and RS denote *Asymmetric punishment*, *Asymmetric endowment*, *Poor strong*, and *Rich strong*, respectively.

*.05 < $p \leq .10$. **.01 < $p \leq .05$. *** $p \leq .01$.

reduction due to punishment. The regression results are shown in Table 4. Including all members from all treatments, we find that, in line with all previous studies, negative deviation from average cooperation is the strongest predictor of punishment received (i.e., it has the largest coefficient and is significant at $p < .01$ in all treatments, F -test). Further, the lower the average group members' cooperation is, the higher is the punishment received under *Symmetry*, *Asymmetric punishment*, and *Rich strong*.²⁸

Next we use the same specification to estimate models for rich and poor types across the treatments with endowment heterogeneity and for strong and weak (punisher) types across the treatments with heterogeneity in punishment effectiveness. In all treatments and for all types, a

negative deviation from group members' cooperation predicts punishment received (at $p < .02$). Across the endowment heterogeneity dimension, the level of cooperation influences punishment received by the rich and strong punisher members in *Rich strong* ($p = .02$) and by the poor in both joint heterogeneity treatments (both at $p < .01$). Across the punishment heterogeneity dimension, the level of cooperation matters for strong punisher members in the joint heterogeneity treatments (at $p < .02$), for weak punisher members in *Asymmetric punishment* ($p = .04$), and for poor and weak punisher members in *Rich strong* ($p < .01$). Let us summarize.

RESULT 4 (INDIVIDUAL PUNISHMENT RECEIVED PER TYPE):

For all types (rich, poor; strong, weak), punishment received is predicted by negative deviations from the average cooperation of the other group members. Lower cooperation levels of the other members increases punishment received by rich and strong punisher and poor and weak

28. It is also marginally significant in *Asymmetric endowment* ($p = .077$, F -test).

TABLE 4
Tobit Model Estimation (Punishment Received)

Treatments Members	ALL	With Endowment Heterogeneity (AE, PS, RS)		With Punishment Heterogeneity (AP, PS, RS)	
	(All Types)	(Rich Type)	(Poor Type)	(Strong Type)	(Weak Type)
Average group cooperation	−4.85*** (1.54)	−1.46 (2.08)	−3.03 (1.93)	−3.35 (2.30)	−3.58** (1.73)
Average group cooperation × AP	0.72 (2.30)	−	−	−	−
Average group cooperation × AE	2.36 (2.08)	−	−	−	−
Average group cooperation × PS	2.53 (2.31)	0.06 (2.96)	−1.97 (2.48)	−2.14 (3.07)	2.15 (2.67)
Average group cooperation × RS	0.16 (2.20)	−2.92 (2.78)	−3.12 (2.52)	−1.36 (3.04)	−2.55 (2.52)
Negative absolute deviation	16.55*** (1.84)	13.19*** (2.50)	10.08*** (2.25)	19.78*** (3.53)	16.55*** (3.98)
Negative absolute deviation × AP	1.58 (3.34)	−	−	−	−
Negative absolute deviation × AE	−3.40 (2.51)	−	−	−	−
Negative absolute deviation × PS	−3.13 (2.62)	3.16 (3.70)	−1.94 (4.00)	−10.98** (4.99)	−0.26 (4.92)
Negative absolute deviation × RS	−1.26 (2.64)	4.50 (3.73)	1.91 (3.10)	−1.56 (4.53)	−4.65 (4.73)
Constant	−3.49** (1.74)	−3.90** (1.75)	−3.52** (1.67)	−5.20*** (1.59)	−4.02*** (1.24)
Treatment dummies	Yes	Yes	Yes	Yes	Yes
Period dummies	Yes	Yes	Yes	Yes	Yes
Observations	8,160	1,740	1,740	1,700	1,700
F(<i>k</i> − 1, <i>N</i> − <i>k</i> − 1)	15.67***	8.00***	7.07***	5.89***	10.71***
Pseudo- <i>R</i> ²	.11	.12	.08	.12	.13

Notes: Dep. variable: “Received punishment” is the number of punishment points received ($\sum P_j$) in period *t*. The table shows the estimated coefficients (standard errors clustered across groups in parentheses). “Average group cooperation” is the average cooperation rate of the other two group members. “Negative absolute deviation” is defined as the negative difference between the rate of the individual’s cooperation and the cooperation rate of the remaining two members. AP, AE, PS, and RS denote *Asymmetric punishment*, *Asymmetric endowment*, *Poor strong*, and *Rich strong*, respectively.

*.05 < *p* ≤ .10. **.01 < *p* ≤ .05. ****p* ≤ .01.

punisher members in the Rich strong treatment and by poor and strong punisher members in the Poor strong treatment.

The result that punishment received is always (and most considerably) related to negative deviations from the average level of cooperation is consistent with the enforcement of the *equal cooperation* norm. This is very much in line with Reuben and Riedl’s (2013) findings on endowment heterogeneity.

V. DISCUSSION AND CONCLUSIONS

We study the effect of joint heterogeneities in endowment and punishment effectiveness on cooperation in a public good game. Our study is motivated by the apparent discrepancy between field situations and laboratory experiments. In the field, heterogeneities in wealth and punishment

power are congruent, while laboratory experiments, so far, have only considered one heterogeneity at a time. The question we set out to answer is *would peer punishment enhance cooperation even in a situation where the rich party (gaining less from the public good than the others) is also the most effective punisher?* The answer to this question is a resounding “yes.” At the bottom line, our study provides additional evidence for the striking effectiveness of peer punishment in facilitating cooperation in a linear public good game.

More precisely, our findings can be summarized as follows. First, in line with Nikiforakis, Normann, and Wallace (2010) and Dorrrough, Glöckner, and Lee (2017), heterogeneity in punishment effectiveness alone does not lead to different cooperation than under symmetry. This result is in line with conjecture Ho(1). Second, at odds with our conjecture Ho(2), endowment

heterogeneity (with equal punishment effectiveness) reduces cooperation compared to symmetry. This stems from the lower cooperation of the rich members. Third, we observe that joint heterogeneities in endowment and punishment effectiveness do not yield lower cooperation than under symmetry. The finding that when the poor are the most effective punishers (and hence can “force” the rich into cooperation), cooperation is higher than under endowment heterogeneity is in line with conjecture Ho(3). By contrast, the result that when the rich are the most effective punishers, cooperation is higher than under endowment heterogeneity is at odds with Ho(4). Note that in the *Asymmetric endowment* and the joint heterogeneities treatments, the rich gain less from cooperation (and stand to lose more if they cooperate and the others defect) than their poorer counterparts. A likely explanation for the high cooperation under congruent heterogeneities is that in this *Rich strong* treatment the poor gain more from cooperation than the rich and thus they have incentive to cooperate. Yet, the rich who are now the most effective punishers feel more confident to cooperate than under endowment heterogeneity alone. Consequently, the equivalence in cooperation between *Poor strong* and *Rich strong* contradicts conjecture Ho(5). However, in *Poor strong*, cooperation is enforced by using more punishment than in *Rich strong*, indicating that congruent heterogeneities could actually be efficiency-increasing.

The negative effect of endowment heterogeneity on cooperation is in line with most linear public good game studies *without peer punishment* (see the meta-analysis by Zelmer 2003, and also Cherry, Kroll, and Shogren 2005; Anderson, Mellor, and Milyo 2008; Keser et al. 2013; Weng and Carlsson 2015; and Hargreaves Heap, Ramalingam, and Stoddard 2016). Moreover, our results of lower cooperation of rich than of poor members are in line with Buckley and Croson (2006). However, to our knowledge, the current study is the first to indicate that endowment heterogeneity reduces cooperation *when parties can punish their peers* (Visser and Burns 2015 and Prediger 2011 find a positive effect, whereas no significant effect is indicated by Weng and Carlsson 2015). One plausible explanation for the ambiguous findings of these studies is the notable differences in participant pools. Herrmann, Thöni, and Gächter (2008) show that peer punishment has a very different effect on cooperation in different countries and cultures. These

authors identify an interesting cultural dimension that affects punishment behavior: the “rule of law.” It is based on “a host of different variables that measure the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, the police, and the courts, as well as the likelihood of crime and violence” (Herrmann, Thöni, and Gächter 2008, 1366). While this variable is very high for Germans, both Prediger (2011, 33) and Visser and Burns (2015, 96) assess such a variable to be rather low for their participant pools. Supporting evidence for the differences in cooperation norms between the participant pools in the latter two treatments and our study is that the average cooperation under their benchmark symmetry (with punishment) treatments is considerably lower than in this current study.²⁹ Even though in these studies cooperation under endowment heterogeneity is higher than under symmetry, their average cooperation under endowment heterogeneity is nevertheless lower than in any of the current study’s treatments. In this regard, the moderating effect of culture is particularly interesting in an experiment with heterogeneities, which creates normative conflicts that may be overcome differently, depending on the prevailing norm in the given society (i.e., what is “fair”? equal amounts contributed? or equal rates of cooperation?). This should be further addressed in future research.

Another issue is that the workhorse in the current study is the linear public good game, an environment in which peer punishment is known to be especially effective in facilitating cooperation. In this respect, using a common pool resource game, Ostrom, Walker, and Gardner (1992) observe that peer punishment alone (i.e., without communication among group members) does not enhance cooperation. Moreover, Cason and Gangadharan (2015) find that the impact of punishment is weaker and takes longer to effect in a nonlinear public good game and a common pool resource game than in a linear public good game. In these games the Nash equilibria and the socially efficient outcomes are less

29. In particular, average cooperation rates are 0.44 in Prediger (2011) and between 0.49 and 0.43 in Visser and Burns (2015). For comparison, average cooperation rates are 0.81 and 0.69 in Reuben and Riedl (2013) and Weng and Carlsson (2015), respectively. Moreover, in both Prediger (2011) and Visser and Burns (2015), average cooperation rates *without punishment* are relatively high (0.47, and between 0.46 and 0.40, respectively).

obvious than in a linear public good game, making it harder to identify free riding and cooperation (see Cason and Gangadharan 2015, 69–72). Furthermore, heterogeneities may bring about normative conflicts, making it harder to coordinate on the socially efficient outcome (see Niki-forakis, Noussair, and Wilkening 2012; Reuben and Riedl 2013). So, another open question for future research is whether our results would hold in other social dilemma situations (e.g., how do congruent heterogeneities affect cooperation in nonlinear social dilemmas?).

In conclusion, relating to the motivation referred to at the beginning of the paper, we believe that the correct interpretation of our laboratory results is not that bilateral punishment would be successful in increasing local or global public good provision in the field under congruent heterogeneities. Numerous factors are different between the stylized public good game that we have employed and even the simplest dilemma in a small-scale society (and even more in a global dilemma, where players represent countries). Yet, our results suggest that congruent heterogeneities in wealth and punishment effectiveness do not necessarily hinder cooperation per se, even in a design where the rich parties stand to gain less from public good provision than their poor counterparts.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.
Appendix S1. Further Tables and Analysis